TEACHING PORTFOLIO

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1. Introduction

Teaching is not merely much more than knowing: it belongs to a *different* category. For any piece of knowledge, any statement, any principle—however elementary—there was a time when we did not understand it. And between not understanding and understanding, something happened.

There was something we saw that turned the confusing into the clear, perhaps even into the obvious. To explain necessarily means stepping down from the place of the obvious and standing on a level footing with the learner, in order to reconstruct that something that lies there, at the centre.

It is a matter of *refreshing intuition*. To understand something is to turn it into something intuitive. To teach it is to offer simple paths that make it so.

Whatever the level, what I care about most is the bond between teacher and student: the concrete, real-time work of reading what a group has and has not understood, and of building the small models and images that make an apparently difficult idea intuitive. I put particular effort into clear exposition and into treating the blackboard as a narrative object, so that a course reads as a single connected story rather than as a list of results. This approach is reflected in consistently strong student evaluations—an overall rating of *very good* across more than 300 documented responses (see Section 8)—and in expository talks, both pedagogical and research-oriented, that have been very well received.

My contact with mathematics and with teaching goes back a long way. I began competing in the Uruguayan National Mathematical Olympiad at the age of twelve, and from fifteen I volunteered as a workshop and seminar instructor for secondary-school students (2010–2015), with an emphasis on problem-solving strategies. In 2013 I took my first formal teaching post at Escuela y Liceo Elbio Fernández, in charge of its Olympiad delegation. In March 2015 I joined IMERL as a Teaching Assistant, beginning the university teaching career detailed below.

Something of how I understand mathematics—and its communication—can also be seen in a few public-facing pieces: an interview centred on teaching (*Mis Profes en FIng*, video), an expository research talk at IMPA (video), and two general-audience mathematics columns (#3, #6).

Current appointment. Faculty member at IMERL since 2015: Teaching Assistant from 2015, and Junior Professor (tenured since 2023) from 2019, with admission to the Full-Dedication Regime approved in December 2025. Junior Professor and Teaching Assistant at CMAT (Faculty of Sciences) in various periods. Currently on leave for a FAPESP postdoctoral fellowship at IME-USP (São Paulo), while maintaining my affiliation with IMERL and the associated research and training activity.

2. University courses taught

The courses are ordered by *degree of responsibility*. First, those in which I held responsibility for coordination and/or for delivering the lectures; then a documented detail of a representative selection; and finally the recitation (practical) activity.

2.1. Courses with coordination and/or lecturing responsibility. Course coordination (overall responsibility: organization, teaching team, assessment):

- Applications of Linear Algebra. Graduate. IMERL, 2nd sem. 2024.
- Discrete Mathematics. CMAT, 2024.
- Complex Analysis. CMAT, 2024.
- Probability and Statistics. IMERL, 1st sem. 2023.
- Multivariable Calculus. IMERL, 1st sem. 2021.
- Introduction to Topology. CMAT, 1st sem. 2017.
- Geometry of Curves and Surfaces. CMAT, 2nd sem. 2017.

Lecturing (main theory course):

- Vector Calculus. IMERL, 1st sem. 2025.
- Geometry and Linear Algebra 1. IMERL, 2nd sem. 2024.
- Differential and Integral Calculus in Several Variables. IMERL, 2nd sem. 2022.
- Differential and Integral Calculus in Several Variables. IMERL, 1st sem. 2021.

2.2. Documented detail of selected courses. Below I document the objectives, contents, assessment methods and organization of the material for a selection of undergraduate courses at the Faculty of Engineering in which I held lecturing responsibility, together with the graduate course I coordinated. The three undergraduate courses are standard curricular units, with two midterms (worth 40 and 60 points) and a pass-by-coursework-or-exam scheme. In all of them I took part in delivering classes, holding office hours, designing and grading assessments, and giving feedback to students.

Differential and Integral Calculus in Several Variables (IMERL, 2021 and 2022) — lecturer.

- **Objectives.** Consolidate single-variable calculus. Understand and handle differential and integral calculus in several real variables. Develop logical reasoning for problem-solving and the ability to reason by analogy.
- **Contents.** Complex numbers, elementary differential equations, sequences and series in $\mathbb{R}$, improper integrals, topology, continuity and differentiability in $\mathbb{R}^n$ (partial and directional derivatives, the differential, tangent plane, chain rule), multiple integrals (Jordan measurability, Fubini, change of variables).
- **Assessment.** Two midterms (40 and 60 points) under the Faculty's pass/exam scheme.
- **Responsibility and personal contribution.** Lecturer in two editions. Preparation of integrative summaries at mid-term and end of course, and of practice midterms (see Section 6).

Vector Calculus (IMERL, 2025) — lecturer.

- **Objectives.** Master differential and integral calculus on curves and surfaces, handle vector fields, and adapt the techniques to analogous situations, in particular applications to physics.
- **Contents.** Continuity and differentiability of fields, free and constrained extrema (Lagrange), inverse and implicit function theorems, curves and line integrals, gradient and conservative fields, Green's theorem, surfaces and surface integrals, Stokes' theorem (with Faraday's and Ampère's laws), curl and divergence, Gauss's divergence theorem.

- **Assessment.** Two midterms (40 and 60 points) under the Faculty's pass/exam scheme.
- **Responsibility and personal contribution.** Lecturer. The applied orientation of the course (electromagnetism, fluid mechanics) was a guiding thread. Overall rating in the student evaluation survey: mean 4.82/5 over 39 responses (see Section 8).

Geometry and Linear Algebra 1 (IMERL, 2024) — lecturer.

- **Objectives.** Handle the basic algebraic techniques (linear systems, row reduction, matrix algebra), structure linear algebra on the geometric model of free vectors, and build the axiomatic model of a vector space from examples.
- **Contents.** Linear systems and matrices (Gauss, rank, Rouché–Frobenius, inverse), determinants (Cramer), geometry of space (dot and cross products, areas and volumes), vector spaces and subspaces (basis, dimension, change of basis), linear transformations (associated matrix, kernel and image).
- **Assessment.** Two midterms (40 and 60 points) under the Faculty's pass/exam scheme.
- **Responsibility and personal contribution.** Lecturer. A large first-year course. Overall rating in the survey: mean 4.81/5 over 43 responses (see Section 8).

Applications of Linear Algebra (IMERL, 2nd sem. 2024) — graduate; coordination and lecture/recitation.

- **Role and responsibility.** Course coordination and delivery of the recitation, in a combined lecture–recitation format, together with Prof. Marcelo Lanzilotta (instructor of record). A PEDECIBA graduate course, which entailed greater academic autonomy: organization of the material, follow-up of the course, and participation in the design of assessments.
- **Objectives.** Master the Jordan canonical form and its applications to the asymptotic behaviour of sequences and matrix powers; and handle spectral tools (Perron–Frobenius, graph spectra) aimed at modelling in other sciences.
- **Contents.** Jordan canonical form (algebraically closed fields and the general case; sequences and matrix powers); applications to economics (Leontief model); eigenvalues of special matrices (localization in the complex plane, normal and Hermitian matrices); matrices leaving cones invariant; Perron–Frobenius theorem (stochastic matrices, game theory, Google's PageRank algorithm, ranking problems); graphs and adjacency matrices (characteristic polynomial, Sachs' theorem, spectral characterizations).
- **Assessment.** Two midterms (40 and 60 points); exemption with 60 or more; right to a theory–practice exam with 25 or more.
- **Contribution and relevance.** The course connects advanced linear algebra with direct applications to engineering and other sciences (networks, page ranking, population dynamics, economics). The joint work with Marcelo Lanzilotta was fluid, efficient, and planned together.

2.3. Courses with recitation (practical) responsibility.

- 2025, 2nd sem. – Geometry and Linear Algebra 1. IMERL.
- 2024, 2nd sem. – Complex Analysis. CMAT.
- 2024, 1st sem. – Discrete Mathematics. CMAT.
- 2023, 1st sem. – Probability and Statistics. IMERL.
- 2021 and 2020, 2nd sem. – Differential Equations. IMERL.
- 2020, 1st sem. – Vector Calculus. IMERL.
- 2019, 2nd sem. – Differential Equations. IMERL.
- 2019, 1st sem. – Single-variable Calculus. IMERL.
- 2018, 2nd sem. – Discrete Mathematics I. IMERL.

- 2018, 1st sem. – Complex Analysis. IMERL.
- 2017, 2nd sem. – Geometry of Curves and Surfaces (CMAT); Discrete Mathematics I (IMERL).
- 2017, 1st sem. – Introduction to Topology (CMAT); Calculus I (IMERL).
- 2016, 2nd sem. – Mathematics II (Biological Sciences, CMAT); Calculus II (IMERL).
- 2016, 1st sem. – Calculus I. IMERL.
- 2015, 2nd sem. – Discrete Mathematics I. IMERL.
- 2015, 1st sem. – Geometry and Linear Algebra 1. IMERL.

3. Supervision and advanced training

- **Master's thesis co-supervision.** Lara de Assumpção Maffei Pierobon (USP), thesis *Topological horseshoes*, defended in 2026. Co-supervised with Fábio Armando Tal.
- **Mentoring of PhD students.** Weekly meetings with four PhD students from Fábio Armando Tal's group (USP): mathematical guidance and discussion of research topics (since March 2026).
- **Specialized reading seminar — Rotation Theory on Surfaces** (IME-USP, 2026). An informal graduate/research seminar aimed at PhD students and postdocs at USP, which I designed and, in practice, run—syllabus, opening lectures, and organization—jointly with Fábio Armando Tal. It follows the extensive treatise *Théorie de la rotation sur les surfaces* that Pierre-Antoine Guihéneuf developed at IMJ-PRG (Sorbonne Université) in connection with his *habilitation* (HDR). It is the most specialized teaching activity I have lead.
- **Graduate seminar.** *Endomorphism Dynamics*, in support of Juliana Xavier. IMERL, 1st sem. 2025.
- **Teacher-training workshops.** Within the *Talleres de Matemática UdelaR* (TMU), co-responsible for problem-solving workshops aimed at primary- and secondary-school *teachers* (2017–2021).

4. Other teaching

- Minicourse *Topics in Topological Plane Dynamics*, with Juliana Xavier. VII Uruguayan Mathematics Colloquium, 2019.
- Systematic Olympiad training (secondary level), at Escuela y Liceo Elbio Fernández and at Monte VI / Los Pinos (2013–2018).

5. Publicly available teaching materials

- **Talleres de Matemática UdelaR (TMU).** Co-responsible for weekly seminars and for preparing dedicated material across five annual editions (2017–2021). More than 30 notes, each with a brief theoretical exposition and a selection of problems. Material available at tmu.edu.uy/seminarios.
- **Minicourse notes** *Topics in Topological Plane Dynamics* (with Juliana Xavier), VII Uruguayan Mathematics Colloquium, 2019. Available here.

6. Objectives and methods

Even though, as a Teaching Assistant or Junior Professor, one's ability to influence the methodology and delivery of a course is limited—all the more so in the early semesters, where the curriculum carries decisive weight—I implemented specific dynamics in the groups under my responsibility to strengthen what reaches the students:

- Two classes per semester devoted to an integrative summary of each half of the course, prepared for that purpose. It is essential to convey that a course is a story, that its content is interconnected and forms a whole.
- Preparation and running of practice midterms in first-year courses, with exercises designed for the occasion.
- Applications of the content inspired by extracurricular literature and popular-science books, in the recitation of Discrete Mathematics I (2015–2018).
- A conviction that the design of assessments should be handled by a small team with a holistic view of the course, as opposed to the automatic assembly of exercises from different areas. I have held this position in the courses I take part in and have been part of that team, both as a representative of the recitation instructors and of the lecturers.

7. Participation in new curricular programs

I took part in the first editions of the redesigns of the Calculus I, Calculus II and Calculus III courses. Even without assigned tasks of coordination or note-writing, I played an active role in readapting the approach to the content for the groups under my responsibility.

8. Student evaluation surveys

The official reports of the Teaching Unit (survey F5, UEFI records) are available. The overall judgement of the instructor (question 11) was consistently **VERY GOOD**. The table below summarizes the documented editions.

Period	**Course**	**Responses**	**Overall (mean)**
2017 – 1st sem.	Calculus 1	86	4.45
2018 – 1st sem.	Complex Analysis	49	4.63
2021 – 1st sem.	Differential and Integral Calculus in Several Variables	19	4.80
2021 – 2nd sem.	Introduction to Differential Equations	23	4.70
2022 – 2nd sem.	Differential and Integral Calculus in Several Variables	4	4.50
2023 – 1st sem.	Probability and Statistics	16	4.20
2024 – 1st sem.	Differential and Integral Calculus in Several Variables	13	4.30
2024 – 2nd sem.	Geometry and Linear Algebra 1	43	4.81
2025 – 1st sem.	Vector Calculus	39	4.82
2025 – 2nd sem.	Geometry and Linear Algebra 1	11	4.91

In all, more than 300 documented student responses, with a weighted mean of the overall judgement around 4.62/5. The three most recent editions—including Geometry and Linear Algebra 1 (2024) and Vector Calculus (2025), both taught as lecturer—average 4.83/5 over 93 responses.

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